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DIGITAL INPUTS MODULE VITAL (DIM-V)

Thales On-Board Platform (TOP)
Functional Verification Test Report (FVTR)

THALES GROUP INTERNAL

Entity Identifier	Document Identifier	DTC	Revision	Customer ID with Revision
15900~9950	7HA-02941-AAAA-980EN	980	009	N/A
Legacy Identifier: *N/A*				

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Template: 83410477-DOC-CAN-EN-009

Follow-up of the evolutions

LOG OF CHANGES				
Revision	**Date**	**ECN #**	**Author**	**Modification**
As 7HA-02919-AAAA-980EN				
001	**2012-07-23**		**S. D'Alimonte**	**First release**
Number changed to 7HA-02941-AAAA-980EN				
001	**2012-10-12**		**S. D'Alimonte**	**Change part number** **Correct requirement tags (Comment 1382190)**
002	**2012-11-29**		**W. Sheng**	**Released test report**
003	**2013-09-27**		**W. Sheng**	**Test report for bug-fix**
004	**2014-02-14**		**S. D'Alimonte**	**Add details of safety components**
005	**2014-10-07**		**W. Sheng**	**Multiple updates**
006	**2017-07-14**		**A. Feng**	**Update with variant ACAA and BCAA test results**
007	**2017-12-14**		**I. Shuryghin**	**Update with variants ABAA and BBAA test results**
008	**2020-12-14**		**I. Shuryghin**	**Traceability Updates**
009	**2020-12-23**		**I. Shuryghin**	**Traceability Updates. *S* FVTR_SAF_04 update.**

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APPROVAL				
	Name	**Role**	**Date**	**Signature**
Written by	I. Shuryghin	CID	2021-03-02	Confirmed via CMCS
Verified by	S. D'Alimonte	HwARC	2021-03-08	Confirmed via CMCS
Approved by	G. Ilie	HEM	2021-03-08	Confirmed via CMCS
Publications				
Approved by the customer if necessary				

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1 Introduction

1.1 Purpose

The document describes the functional verification of the item DIM. In the first Edition it specifies the tests. In the following Editions it describes the results of all functional tests, which are carried out during the functional test verification.

The item is part of the system TOP system.

1.2 Intended audience

The intended audience of this document are the following functions of the TOP:

- Development team
- Safety manager (SAF)
- Assessor (AS)
- Validation (VAL)

1.3 Evolution of the document

The document is first issued during detailed design phase, and will be updated on feature or requirement changes, if necessary.

For type and reasons of changes please refer to section "Editions and Change History" at the beginning of this document.

1.4 Configuration management

This document is under configuration control in CMCS. The document is to be released by the corresponding release authority, defined in the documentation plan [DOP] of the project. The final version will be archived in PALMA.

1.5 References

The mentioned edition is the one used for creation of the document. It is not intended to update the document, if the edition of the reference document is changed.

1.5.1 Referenced documents

Table 1: Referenced documents

Doc-ID	Document number	ED	Title
[DOP]	7HA-02990-0001-931EN	07P07	Documentation Plan TOP

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Doc-ID	Document number	ED	Title
[HWADD_TOP]	7HA-02990-0018-434EN	05	TOP HW Architecture Design Document
[INSUCO_TOP]	7HA-02990-0019-959EN	02P01	TOP Insulation Coordination
[HW-IRS_DIM_VI]	7HA-02919-AAAA-306EN	03	HW Item Requirements Specification Digital Inputs Module Vital
[EBOM_DIM_VI]	7HA-02941-BAAA-001EN 7HA-02941-AAAA-001EN 7HA-02941-ACAA-001EN 7HA-02941-BCAA-001EN 7HA-02941-ABAA-001EN 7HA-02941-BBAA-001EN	10.314. 3 02.4 02.4 01.14 01.7	Parts List (EBOM) Digital Inputs Module Vital +4 Parts List (EBOM) Digital Inputs Module Vital Parts List (EBOM) Digital Inputs Module Vital Parts List (EBOM) Digital Inputs Module Vital +4 Parts List (EBOM) Digital Inputs Module Vital Parts List (EBOM) Digital Inputs Module Vital +4
[IID_DIM_VI]	7HA-02941-AAAA-576EN	03	Assembly Drawing Digital Inputs Module Vital
[RR-PBA_DIM_VI]	7HA-02941-AAAA-909EN 7HA-02941-BAAA-909EN	04 03	Reliability Report Digital Inputs Module Vital
[CD_DIM_VI]	7HA-02941-AAAA-030EN	05	Circuit Diagram Digital Inputs Module Vital
[PBATD_DIM_VI]	7HA-02941-AAAA-040EN 7HA-02941-AAAA-400EN 7HA-02941-AAAA-581EN	05 05 09	Printed Board Assembly Document Set (Assembly Drawing, PBA Test Data, Design Project File) Digital Inputs Module Vital
[DERATING]	8DC-01043-0001-521EN	01	Component Derating Requirements

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Doc-ID	Document number	ED	Title
[TOP_FMDD]	3CU-04016-0031-549EN	06P01	TOP Functional Module Design and Requirements Document
[QTR]	7HA-02950-AAAA-279EN	01	QTR for TOP Subracks (6U and 3U)

1.5.2 Additional references

Table 2: Additional references

Doc-ID	Document number	Title
[Gloss]	8DC-10110-9000-WDAFA	Definitions, Terms, Abbreviations
[SAC_GUIDE]	8AK-00070-0090-955EN	Guidelines and constraints for the usage of inherent fail safe components
[SAC_MELF0204]	1AB139840001-905EN	Safety Application Conditions and Process Adherence (MELF0204)

1.5.3 Standards

Table 3: Standards

Doc-ID	Publisher	Date	Title
EN 50129	CENELEC	2003	Railway applications - Communication, signalling and processing systems - Safety related electronic systems for signalling
1474.1	IEEE	2004	Standard for Communications-Based Train Control (CBTC) Performance and Functional Requirements
AREMA	AREMA	2010	Communications & Signals Manual of Recommended Practice.

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1.6 Definitions and abbreviations

1.6.1 Definitions

All definitions, used in this document, are determined in [Gloss], except:

Table 4: Document specific definitions

Term	Description
HP (Horizontal pitch)	Measure for printed board front panel width (1 HP = 5,08mm)
U (Unit)	Measure for mounting rack height (1U = 44,45mm)

1.6.2 Abbreviations

All abbreviations used in this document are explained in [Gloss], except:

Table 5: Document specific abbreviations

Abbreviation	Meaning
AC	Alternating Current
AREMA	American Railway Engineering and Maintenance of Way Association
BOM	Bill of Material
CAD	Computer Aided Design
CD	Circuit Diagram
CMCS	Configuration Management on Client / Server
DC	Direct Current
DIM	Digital Input Module
DIM-VI	Digital Input Module – Vital
EBOM	Electrical Bill of Materials
EEPROM	Electrically Erasable Programmable Read Only Memory
ESD	Electrostatic Discharge
HP	Horizontal Pitch (1 HP = 0.2” or 5.08 mm)
HW	Hardware
I^2C	Inter Integrated Circuit Bus

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Abbreviation	Meaning
IEEE	Institute of Electrical and Electronic Engineering
IO	Input Output
LED	Light Emitting Diode
LFD	Latent Failure Detection
MTBF	Mean Time Between Failure
MTTR	Mean Time To Repair
PBA	Printed Board Assembly
PSU	Power Supply Unit
RoHS	Reduction of Hazardous Substances
SW	Software
TBD	To Be Determined
TOP	Thales Onboard Platform
U	Rack Unit (1 U = 1.75” or 44.45 mm)
UUT	Unit Under Test
WEEE	Waste Electrical and Electronic Equipment

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2 Summary

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Result: passed 55
Test case: *S* FVTR_SAF_04 55
Test: Inherent fail-safe components SAC usage validation 55
Result: passed 56
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Result: passed 61

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3 Summary of ClearQuest tickets

Table 6: List of ClearQuest tickets

None

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4 Test Assembly for Verification

4.1 Test item

Table 7: Test Item

ID	Description	Item number	Ed.	Manufacturing number
1	DIM Board (DIM V +4) (12 x 96V/110V I/P + 4 x 24V FAST I/P, Input voltage: 67.2 – 137.5 V DC + 24 V DC)	7HA 02941 BAAA (part of HU 7HA02919BAAA)	BOM 10.3 CD: 05	EG124807687
2	DIM Board (DIM Vital) (16 x 96V/110V I/P; Input voltage: 67.2 – 137.5 V DC)	7HA 02941 AAAA (part of HU 7HA02919AAAA)	BOM 14.3 CD: 05	EG124807642
3	DIM Board (DIM Vital) (16 x 24V / 36V I/P; Input voltage: 16.8 – 45 V DC)	7HA 02941 ACAA (part of HU 7HA02919ACAA)	BOM 02.4 CD: 05	EG164690250
4	DIM Board (DIM V +4) (12 x 24V / 36V I/P + 4 x 24V FAST I/P; Input voltage: 16.8 – 45 V DC + 24 V DC)	7HA 02941 BCAA (part of HU 7HA02919BCAA)	BOM 02.4 CD: 05	EG164690298
5	DIM Board (DIM Vital) (16 x 48V / 72V I/P; Input voltage: 33.6 – 90 V DC)	7HA 02941 ABAA (part of HU 7HA02919ABAA)	BOM 1.14 CD: 05	EG172390925, EG172390931, EG172390933
6	DIM Board (DIM Vital) (12 x 48V / 72V I/P + 4 x 24V FAST I/P; Input voltage: 33.6 – 90 V DC + 24 V DC)	7HA 02941 BBAA (part of HU 7HA02919BBAA)	BOM 1.7 CD: 05	EG173590721

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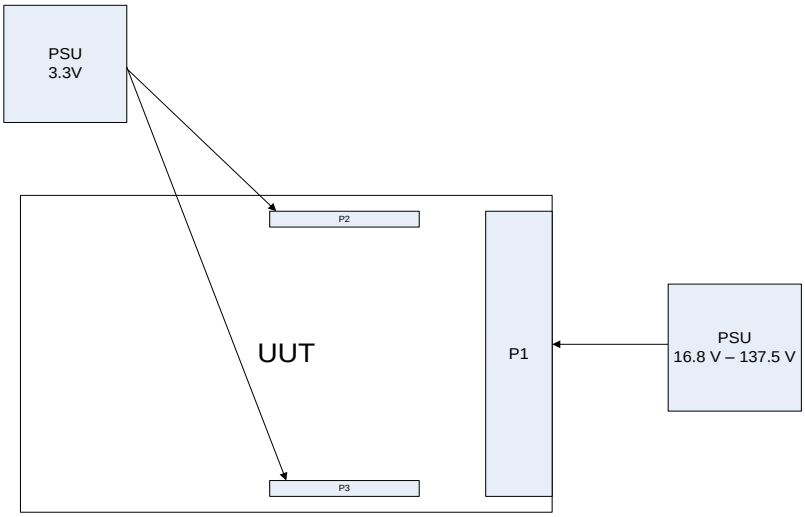
4.2 Test configuration and definitions

4.2.1 Test environment

Test Lab

4.2.2 Physical test configuration

Figure 1: Test Setup



4.2.3 Test software

None Required

4.2.4 Test description

See test procedures

4.2.5 Failure criteria

See test procedures

4.3 Test Equipment

4.3.1 Calibrated test equipment

The following calibrated test equipment was used:

Table 8: Calibrated Test Equipment

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Entity Identifier	Document Identifier	DTC	Revision	Customer ID with Revision
15900~9950	7HA-02941-AAAA-980EN	980	009	N/A
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ID	Measurement Device	Type	Inventory Number	Calibration Date	Calibration Due
1	Meter	Fluke 179	TEO-0377	2013-03-26	2014-03-26
2	Scope	Tektronix TDS 2024B	TEO-0365	2013-03-28	2014-03-28
3	Hi-Pot Tester	AV/DV-50VA-20/10V	TEO-0257	2013-04-23	2014-04-23
4	Multimeter	Fluke 77	TE0-0047	2017-03-28	2018-03-28
5	Oscilloscope	Tektronix TDS2012 + 2 P2220 probes	TE0-0283 T92-0113 T92-0115	2016-09-27	2017-09-27
6	Multimeter	Fluke 289	TE0-0447	2016-09-29	2017-09-29
7	Oscilloscope	Tektronix TDS2012	TE0-0324	2016-09-27	2017-09-27
8	Multimeter	Fluke 289	TE0-0446	2017-09-27	2019-09-27
9	Oscilloscope	Tektronix TDS2012	TE0-0326	2017-09-26	2018-09-26

4.3.2 Non-calibrated test equipment

Table 9: Non-calibrated Test Equipment

ID	Measurement Device	Type	Inventory Number
1	Power Supply	KIKUSUI PAD 70-8L 0~70V 8A	TEO-0220
2	Power Supply	PAN 110-10A	TEO-0558
3	Power Supply	BK Precision 1761	T91-0049
4	Function Generator	HP 3310B	TEO-0042
5	Signal generator	BK Precision 4011A	TE0-0375
6	Power Supply	Kikusui PAS80-4.5	TE0-0560
7	Power supply	BK Precision 9110	PS-0001
8	Power supply	GwInstek GPS-3303	TEO-0545

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4.3.3 Test software

None

4.4 Explanation concerning the following tests

Unless otherwise specified, for the subsequent tests, the signal and component names below refer to the circuit diagram [CD_DIM_VI] of DIM.

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5 Verification of mechanical characteristics

Test case:	**FVTR_MECH_01**
Test:	Labelling and Assembly
Description:	Examination of correct labelling and assembly of parts according to [IID_DIM_VI] and [PBATD_DIM_VI].
Failure criteria:	Labelling or assembly of parts is incorrect.
Reference:	Standard Test
Date:	2017-10-24
Test carried out by:	I. Shuryghin
Result:	**passed**
Comment:	Checked by visual inspection - ESD warning label and item barcode label are placed; - All components are installed.

[FVTR_DIM_VI_0002]

Test case:	**FVTR_MECH_02**
Test:	Surfaces
Description:	Visual examination of surfaces for damage and condition (eg. Bubbles in conformal coating). Check that there are no burrs or sharp edges.
Failure criteria:	Damaged surfaces or sharp edges
Reference:	Standard Test, HW-IRS_DIM_VI_485
Date:	2017-10-24
Test carried out by:	I. Shuryghin
Result:	**passed**

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Comment: Checked by visual inspection, no damage surface or sharp edge is found.

[HW-IRS_DIM_VI_485]

[FVTR_DIM_VI_0003]

Test case: **FVTR_MECH_03**

Test: Coding

Description: Coding pegs shall be installed in the bottom handle of the face-plate as defined in [HWADD_TOP].

Failure criteria: Coding pegs missing or incorrectly installed.

Reference: HW-IRS_DIM_VI_425, HW-IRS_DIM_VI_479

Date: 2017-12-14

Test carried out by: I. Shuryghin

Result: **passed**

Comment: Sample boards are checked according to BOMs. Coding D4 E1 F1 is used for variants AAAA, ABAA, and ACAA; D2 E1 F1 - for variants BAAA, BBAA, and BCAA.

[HW-IRS_DIM_VI_425][HW-IRS_DIM_VI_479]

Test case: **FVTR_MECH_04**

Test: Adjustable parts

Description: All connectors and switches have to be operable without excessive force

Failure criteria: Operation only possible with additional equipment.

Reference: Standard Test

Date: 2017-11-03

Test carried out by: I. Shuryghin

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Result: **passed**

Comment: There is no switch used on the board.
The board to board connectors are tested by plug-in and removal of a daughter board and the back connector is tested by inserting and removing the board assembly from the subrack slot. No excessive force is required for connectors operation.

[FVTR_DIM_VI_0005]

Test case: **FVTR_MECH_05**

Test: ESD Strip

Description: The ESD strip on the printed board consists of two separate segments, which are not connected with each other. One segment reaches the front plate; the other one reaches the rear end of the board and is connected with electronic ground via a resistor. The gap between the segments is larger than the contact slider of the subrack.

Failure criteria: ESD strip is not as described.

Reference: HW-IRS_DIM_VI_534

Date: 2017-10-24

Test carried out by: I. Shuryghin

Result: **passed**

Comment: Strip examined and measured.
The gap is 12mm, larger than 2mm which is the contact slider width.

[HW-IRS_DIM_VI_534]

[FVTR_DIM_VI_0006]

Test case: **FVTR_MECH_06**

Test: Plastic Cover of Assembly Bottom

Description: No plastics cover should be included on the assembly, due to fire hazards.

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Failure criteria: Cover installed

Reference: HW-IRS_DIM_VI_472

Date: 2017-10-24

Test carried out by: I. Shuryghin

Result: **passed**

Comment: Plastic cover is NOT installed on solder side.

[HW-IRS_DIM_VI_472]

Test case: **FVTR_MECH_07**

Test: Locking Varnish

Description: The screws holding the PCB must be secured with locking varnish.

Failure criteria: Missing locking varnish.

Reference: Standard Test

Date: 2017-10-24

Test carried out by: I. Shuryghin

Result: **passed**

Comment: Varnish for all variants is specified in mechanical drawing 7HA02919AAAA_040EN and in all HU BOMs with F/N 19.

[FVTR_DIM_VI_0007]

Test case: **FVTR_MECH_08**

Test: Variants

Description: The following variants shall be defined:

Variant	Description

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1.7HA 02941 ACAA (24V / 36V I/P)	8 vital or 16 non-vital 24 or 36 V DC inputs
2.7HA 02941 ABAA (48V / 72V I/P)	8 vital or 16 non-vital 48 or 72 V DC inputs
3.7HA 02941 AAAA (96V / 110V I/P)	8 vital or 16 non-vital 96 or 110 V DC inputs
4.7HA 02941 BCAA (12 x 24V / 36V I/P + 4 x 24V FAST I/P)	6 vital or 12 non-vital 24 or 36 V DC inputs + 4 24 V DC fast semiconductor inputs
5.7HA 02941 BBAA (12 x 48V/72V I/P + 4 x 24V FAST I/P)	6 vital or 12 non-vital 48 or 72 V DC inputs + 4 24 V DC fast semiconductor inputs
6.7HA 02941 BAAA (12 x 96V/110V I/P + 4 x 24V FAST I/P)	6 vital or 12 non-vital 96 or 110 V DC inputs + 4 24 V DC fast semiconductor inputs

All variants shall share the same layout and be simple BOM variants.

Failure criteria: Available variants do not match the above requirements.

Reference: HW-IRS_DIM_VI_392, HW-IRS_DIM_VI_393, HW-IRS_DIM_VI_394, HW-IRS_DIM_VI_400, HW-IRS_DIM_VI_401, HW-IRS_DIM_VI_402, HW-IRS_DIM_VI_403, HW-IRS_DIM_VI_404, HW-IRS_DIM_VI_405

Date: 2017-10-24

Test carried out by: I. Shuryghin

Result: **passed**

Comment: All boards have been builtbased on the same bare board according to variant table in [CD_DIM_VI]. Variant 4, 5 and 6 have 4 24V fast semiconductor inputs and 6 vital (12 non-vital) metallic interface inputs. The difference between 4, 5 and 6 is the maximum voltage for the metallic inputs.

[HW-IRS_DIM_VI_392][HW-IRS_DIM_VI_393][HW-IRS_DIM_VI_394][HW-IRS_DIM_VI_400][HW-IRS_DIM_VI_401][HW-IRS_DIM_VI_402][HW-IRS_DIM_VI_403][HW-IRS_DIM_VI_404][HW-IRS_DIM_VI_405]

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[FVTR_DIM_VI_0008]

Test case:	**FVTR_MECH_09**
Test:	Dimensions
Description:	The board shall be a standard 3U Eurocard (160 x 100 mm) with a width of 6 HP.
Failure criteria:	The above dimensions are not met.
Reference:	HW-IRS_DIM_VI_417, HW-IRS_DIM_VI_418
Date:	2017-10-24
Test carried out by:	I.Shuryghin
Result:	**passed**
Comment:	The printed board size is 160 x 100 mm, specified in bare board drawing 7HA029330024-003EN. The assembly width is 6HP, fixed by the front plate 3JA690040072, F/N 60 in HU BOM.

[HW-IRS_DIM_VI_417][HW-IRS_DIM_VI_418]

[FVTR_DIM_VI_0009]

Test case:	**FVTR_MECH_10**
Test:	Faceplate
Description:	The layout of the faceplate elements shall match [HWADD_TOP].
Failure criteria:	The faceplate does not match.
Reference:	HW-IRS_DIM_VI_423
Date:	2017-10-24
Test carried out by:	I. Shuryghin
Result:	**passed**
Comment:	Faceplate elements are checked, matching with [HWADD_TOP] described: two status LEDs for replica 1 on upper portion, and two status LEDs for replica 2 on lower portion.

[HW-IRS_DIM_VI_423]

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[FVTR_DIM_VI_0010]

Test case:	**FVTR_MECH_11**
Test:	Daughterboard Fit
Description:	The daughterboard must fit correctly with all connectors fully mated.
Failure criteria:	Daughterboard does not fit correctly.
Reference:	HW-IRS_DIM_VI_422
Date:	2017-11-03
Test carried out by:	I. Shuryghin
Result:	**passed**
Comment:	The daughterboard connectors were fully mated with the main board connectors, and there was no visible gap between connectors.

[HW-IRS_DIM_VI_422]

[FVTR_DIM_VI_0011]

Test case:	**FVTR_MECH_12**
Test:	Slot Compatibility
Description:	The card must fit correctly in the TOP IO subrack with the daughterboard installed. Both the board connector and the daughterboard connector must fully mate with the backplane when the card is inserted.
Failure criteria:	Card does not fit or connectors do not mate correctly.
Reference:	HW-IRS_DIM_416, HW-IRS_DIM_422
Date:	2017-12-06
Test carried out by:	I. Shuryghin
Result:	**passed**
Comment:	Both the main board rear connector and the daughterboard rear connector were fully mated with the backplane, and there was no visible gap between connectors. Test was done with new boards:

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7HA-02919-AAAA Ed.:04 EG 14290 4445 with
7HA-02941-AAAA Ed.:09 EG 14290 4450 and
7HA-02930-AAAA Ed.:07 EG 13411 0781 (Daughterboard)
All variants was built form the same bare boards, connectors, and mechanical parts. AAAA variant can be regarded as representative for all other variants.

[HW-IRS_DIM_VI_416][HW-IRS_DIM_VI_422]

[FVTR_DIM_VI_0012]

Test case:	**FVTR_MECH_13**
Test:	Conformal Coating
Description:	Depending on the board variant the board shall be coated.
Failure criteria:	Coated variant is not coated.
Reference:	HW-IRS_DIM_292
Date:	2017-12-11
Test carried out by:	I. Shuryghin
Result:	**passed**
Comment:	The prototypes are not coated. The design supports coating as specified in assembly drawing 7HA02941AAAA_040EN and in HU BOMs for ABAA and BBAA variantes with F/N 6. BOMs of variants AAAA, ACAA, BAAA, and BCAA don't contain the conformal coating lacquer, as they were issued before the requirement (to include the lacquer in to BOMs) was appeared.

[HW-IRS_DIM_VI_292]

[FVTR_DIM_VI_0013]

Test case:	**FVTR_MECH_14**
Test:	Cooling
Description:	No active cooling mechanisms (fans, peltiers, etc.) shall be used.
Failure criteria:	Active cooling is present.
Reference:	HW-IRS_DIM_427
Date:	2017-10-24

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Test carried out by: I. Shuryghin

Result: **passed**

Comment: No active cooling mechanism is found on the board.

[HW-IRS_DIM_VI_427]

[FVTR_DIM_VI_0014]

Test case: **FVTR_MECH_15**

Test: Component Derating

Description: All components on the DIM board shall have their voltage, power or current stress derated as per Thales Component Derating Requirements [DERATING].

Failure criteria: Derating requirements are not met.

Reference: HW-IRS_DIM_413

Date: 2017-10-24

Test carried out by: I. Shuryghin

Result: **passed**

Comment: Schematic checked: critical parts checked by their voltage stress, current stress, power stress compared with Derating Requirements [DERATING] with regard to their package style.

[HW-IRS_DIM_VI_413]

[FVTR_DIM_VI_0016]

Test case: **FVTR_MECH_17**

Test: Materials

Description: Materials used shall be fire resistant, low smoke, zero halogen and comply with RoHS and WEEE requirements. The use of the following materials shall be avoided or justified:

- polyvinyl chlorides
- asbestos
- cadmium

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- beryllium
- cobalt
- lead
- arsenic
- hexavalent chromates
- zinc
- Bromine flame proof agents
- Mercury

Failure criteria: The above material requirements are not met.

Reference: HW-IRS_DIM_472, HW-IRS_DIM_489, HW-IRS_DIM_490

Date: 2017-05-19

Test carried out by: A. Feng

Result: **passed**

Comment: Material and component relevant properties were successfully verified via Palma.

[HW-IRS_DIM_VI_472][HW-IRS_DIM_VI_489][HW-IRS_DIM_VI_490]

[FVTR_DIM_VI_0017]

Test case: **FVTR_MECH_18**

Test: Surge Protection Devices

Description: Surge protection devices shall not burn, melt or ignite nearby surfaces during operation or surge conditions.

Failure criteria: Burning or melting of the protector or nearby surfaces.

Reference: HW-IRS_DIM_VI_285

Date: 2017-10-

Test carried out by: I. Shuryghin

Result: **passed**

Comment: The surge protection devices, TVS diodes are used together with a fuse for protection. The surge device is further validated during EMC qualification testing.

[HW-IRS_DIM_VI_285]

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6 Verification of Function

[FVTR_DIM_VI_0018]

Test case:	**FVTR_FUNC_01**
Test:	Wiring and Grounding
Description:	Verify the wiring of the DIM according to [CD_DIM_VI]. Verify the resistance between the faceplate and the ESD strip is < 1 Ω.
Failure criteria:	Error in circuit or resistance between ESD strip and faceplate is > 1 Ω.
Reference:	HW-IRS_DIM_492
Date:	2017-11-03
Test carried out by:	I. Shuryghin
Result:	**passed**
Comment:	Measured resistance is 0.3 Ω.

[HW-IRS_DIM_VI_492]

[FVTR_DIM_VI_0019]

Test case:	**FVTR_FUNC_02**
Test:	External Interfaces
Description:	The external interfaces (connectors, types and pinouts) shall meet the requirements of [HW-IRS_DIM_VI] and [HWADD_TOP].
Failure criteria:	Connectors or pinouts do not match.
Reference:	HW-IRS_DIM_419, HW-IRS_DIM_447, HW-IRS_DIM_446, HW-IRS_DIM_448, HW-IRS_DIM_449, HW-IRS_DIM_450, HW-IRS_DIM_451
Date:	2017-10-24
Test carried out by:	I. Shuryghin
Result:	**passed**

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Comment: Connectors and pinouts match the requirements

[HW-IRS_DIM_VI_419][HW-IRS_DIM_VI_447][HW-IRS_DIM_VI_446][HW-IRS_DIM_VI_448][HW-IRS_DIM_VI_449][HW-IRS_DIM_VI_450][HW-IRS_DIM_VI_451]

[FVTR_DIM_VI_0021]

Test case: **FVTR_FUNC_03**

Test: Power Supplies

Description: The Module shall be powered by two sets of 5 V DC and 24 V DC power supplies provided from the backplane, one for each replica.

Failure criteria: The board requires other power supplies, or sharing power for two replicas.

Reference: HW-IRS_DIM_469

Date: 2017-10-24

Test carried out by: I. Shuryghin

Result: **passed**

Comment: DIM module is powered by two sets of 5VDC and 24VDC provided from the backplane, one set for each replica.

[HW-IRS_DIM_VI_469]

[FVTR_DIM_VI_0022]

Test case: **FVTR_FUNC_04**

Test: Power Supply Current

Description: The power consumption of the board shall be less than 8 W. The current drawn from each power rail shall be less than:

Power Rail	Maximum Current
3.3 V	25 mA

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5.0 V	0 mA
24 V	0 mA

Failure criteria: Current or power limits are exceeded.

Reference: HW-IRS_DIM_491, HW-IRS_DIM_470, HW-IRS_DIM_486

Date: 2017-11-03

Test carried out by: I. Shuryghin

Result: **passed**

Comment: Main board power is 3.3VDC which is supplied through the microprocessor daughterboard. See table below for 3.3V current consumption.
For input circuit power consumption which is supplied externally, calculation is based on measurement in FVTR_FUNC_12: maximum 137.5[V] x 3.3[mA] x 16 =7.26[w].
As the sum of the above two, the total power consumption for the board is less than 8w.

Variant	Current consumption on 3.3VA [All inputs active, no LFD enabled]	Current consumption on 3.3VB [All inputs active, no LFD enabled]	Power consumption on 3.3VA and 3.3VB	Note
7HA02941BAAA	7.4 mA	7.4 mA	49mw	Measured. One 96/110V input channel consumes 0.33mA on 3.3V rail; One 24V fast semiconductor input channel consumes 2.66mA on 3.3V rail.

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7HA02941AAAA	2.7 mA	2.7 mA	18mw	Measured
7HA02941ABAA	3.2 mA	3.2 mA	21.1mw	Measured
7HA02941ACAA	6.56 mA	6.64 mA	43.6mw	Measured
7HA02941BBAA	7.72 mA	7.74 mA	51.02mw	Measured
7HA02941BCAA	10.8 mA	10.9 mA	71.6mw	Measured

[HW-IRS_DIM_VI_491][HW-IRS_DIM_VI_470][HW-IRS_DIM_VI_486]

[FVTR_DIM_VI_0023]

Test case: **FVTR_FUNC_05**

Test: Architecture

Description: The DIM shall support 8 checked-redundant vital or 16 non-vital discrete inputs. It shall have 16 separate input circuits, each providing status to both microcontrollers. Vital inputs shall be connected to 2 separate input circuits, while non-vital inputs are connected to only 1 input circuit. A mixture of vital and non-vital signals may be connected to the same card. When combined with the CAN-IO daughterboard the DIM shall provide the status of the inputs over a CAN bus.

Failure criteria: The above architecture is not realized.

Reference: HW-IRS_DIM_371, HW-IRS_DIM_373, HW-IRS_DIM_374, HW-IRS_DIM_484, HW-IRS_DIM_375, HW-IRS_DIM_377, HW-IRS_DIM_378, HW-IRS_DIM_379

Date: 2017-10-24

Test carried out by: I. Shuryghin

Result: **passed**

Comment: Checked schematic, the above mentioned architecture is realized.

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[HW-IRS_DIM_VI_371][HW-IRS_DIM_VI_373][HW-IRS_DIM_VI_374][HW-IRS_DIM_VI_484][HW-IRS_DIM_VI_375][HW-IRS_DIM_VI_377][HW-IRS_DIM_VI_378][HW-IRS_DIM_VI_379]

[FVTR_DIM_VI_0024]

Test case:	**FVTR_FUNC_06**
Test:	Input Circuit Design
Description:	The internal architecture of the DIM shall follow the concept depicted in the [HW-IRS_DIM_VI].
Failure criteria:	The design does not follow the concept in the IRS.
Reference:	HW-IRS_DIM_390
Date:	2017-10-24
Test carried out by:	I. Shuryghin
Result:	**passed**
Comment:	The concept is followed: - two functionality identical circuits can be wired to form a vital input; - two functionality identical circuits can be wired individually to form two non-vital inputs.

[HW-IRS_DIM_VI_390]

[FVTR_DIM_VI_0043]

Test case:	**FVTR_FUNC_07**
Test:	Input Circuit Threshold
Description:	Input Circuits of DIM shall have a voltage threshold type transfer function as protection against invalid DC signals and AC induced Plant noise.
Failure criteria:	The input does not have voltages thresholds as described.
Reference:	HW-IRS_DIM_382
Date:	2017-10-26
Test carried out by:	I. Shuryghin
Result:	**passed**

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Comment: Input circuit was checked, Zener diode is used to set the threshold.

[HW-IRS_DIM_VI_382]

[FVTR_DIM_VI_0044]

Test case: **FVTR_FUNC_08**

Test: Input Filtering

Description: Input Circuits DIM shall have a low pass filter type transfer function as protection against AC induced Plant noise.

Failure criteria: Filtering is not present

Reference: HW-IRS_DIM_383

Date: 2017-10-26

Test carried out by: I. Shuryghin

Result: **passed**

Comment: Low pass filter function is presented. It resulted from inspection of the schematics (7HA-02941-AAAA-030EN) and verified on the actual PCB.

[HW-IRS_DIM_VI_383]

[FVTR_DIM_VI_0045]

Test case: **FVTR_FUNC_09**

Test: Wetting Current

Description: The DIM module shall provide the wetting current (self-cleaning current for the Plant relay contacts) function by implementing the concept described in [TOP_FMDD].

Failure criteria: Wetting current is not provided.

Reference: HW-IRS_DIM_376

Date: 2017-10-26

Test carried out by: I. Shuryghin

Result: **passed**

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Comment: Analysis: the wetting current is provided when LFD control is enabled by the fact that the input impedance/resistance of the circuit is reduced by the short created by the LFD switch, thus the input current drawn is increased.
Test: see FVTR_FUNC_12.

[HW-IRS_DIM_VI_376]

[FVTR_DIM_VI_0046]

Test case: **FVTR_FUNC_10**

Test: Latent Failure Detection

Description: Input circuits shall provide Latent Failure Detection (LFD) test capabilities for protection against component failures, component values changes or any other failures on the board, which can result in the following non self-revealing faults:

a) Input “High” threshold level decrease;

b) Low pass filtering time constant decrease;

c) False input status transition from “Low” to “High”;

d) Status lock-up in “High” state after input transition from “High” to “Low”;

e) Short circuit to any other “High” state circuit on the board.

Failure criteria: Latent failure detection circuits are not present or do not allow detection of the faults listed above.

Reference: HW-IRS_DIM_384

Date: 2012-11-06

Test carried out by: W. Sheng, verified by I. Shuryghin

Result: **passed**

Comment: Schematic analysis found, the faults listed above can be detected:
1. If the Zener threshold decreases, the time response following the LFD switch actuation is reduced because it takes less time for the capacitor to get charged with a voltage higher than the threshold;
2. If the capacitor value decreases (Low pass filter time constant

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decreases), the time response following the LFD switch actuation is reduced because of the reduced value of the capacitor;
3. If the input status is falsely transitioned to high, the time response following the LFD switch actuation is increased to infinite because the status is transitioned to low during LFD actuation and remains low as no energy input to charge the capacitor, so the capacitor voltage is unable to reach the input high threshold;
4. If the status is locked up in “High” state after the input transition from “High” to “Low”, the time response following the LFD switch actuation is reduced to zero because no capacitor charging is required for the status to reach “High”, as it is stuck in “High”.
5. If a short circuit to other “High” state circuit exists, the time response following the LFD switch actuation is reduced to zero because no capacitor charging is required for the status to reach “High”, as it is shorted to a “High” state circuit.

[HW-IRS_DIM_VI_384]

[FVTR_DIM_VI_0025]

Test case: **FVTR_FUNC_12**

Test: Wetting Current

Description: The input current must be as specified in the table below.

Table 10: Wetting Current

Input Board Variant	Battery Voltage [V]			Target Input Current (nom. conditions) [mA]	
	Min. (-30%)	Nom.	Max. (+25%)	LFD ON	LFD OFF
7HA 02941 ACAA	16.8	24	30	20.0	4.0
	25.2	36	45	30.0	7.7
7HA 02941 ABAA	33.6	48	60	15.0	3.0
	50.4	72	90	22.5	5.7
7HA 02941 AAAA	67.2	96	120	10.0	2.0
	77	110	137.5	11.5	2.5

Failure criteria: The current drawn is not as specified.

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Reference: HW-IRS_DIM_VI_376

Date: 2017-12-11

Test carried out by: I. Shuryghin

Result: **passed**

Comment: The targeted current is achieved on the samples boards, see measurements below. Wetting current circuits for variants AAAA, ABAA, and ACAA are absolutely edentical to corresponding circuits for variants BAAA, BBAA, and BCAA. PCB is the same for all variants. Accordingly it is sufficient to make measurements for only variants AAAA, ABAA, and ACAA.

Input Board Variant	Input voltage	One channel input current drawn (no LFD)	One channel input current drawn (with LFD)	Note
7HA 02941 AAAA	96V	1.9 mA	9.7 mA	Pass
	110V	2.4 mA	11.1 mA	Pass
	137.5V	3.3 mA	13.9 mA	For reference only
7HA 02941 ACAA	24V	3.37 mA	11.41 mA	pass
	36V	6.29 mA	17.11 mA	pass
	45V	8.48 mA	21.38 mA	For reference

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				only
7HA 02941 ABAA	48V	3.03 mA	9.87 mA	Pass
	72V	5.58 mA	14.8 mA	Pass
	90V	7.51 mA	18.49 mA	For reference only

[HW-IRS_DIM_VI_376]

[FVTR_DIM_VI_0026]

Test case: **FVTR_FUNC_13**

Test: Input Threshold and Timing

Description: The input thresholds and timing for the normal input circuits shall meet the requirements below.

Table 11: Input Parameters

Variant	Voltage			V_{IH}	V_{IL}	t_{ON}	t_{OFF}
	Min.	Nom.	Max				
7HA 02941 ACAA	16.8 V	24 V	30 V	≤ 15 V	≥ 6 V	1 ≤ t_{ON} ≤ 12 ms	≤ 0.5 ms
	25.2 V	36 V	45 V				
7HA 02941 ABAA	33.6 V	48 V	60 V	≤ 30 V	≥ 12 V	1 ≤ t_{ON} ≤ 12 ms	≤ 0.5 ms
	50.4 V	72 V	90 V				
7HA 02941 AAAA	67.2 V	96 V	120 V	≤ 60 V	≥ 24 V	1 ≤ t_{ON} ≤ 12 ms	≤ 0.5 ms
	77 V	110 V	137.5 V				

Failure criteria: The input thresholds or timing are not met.

Reference: HW-IRS_DIM_VI_395, HW-IRS_DIM_VI_396

Date: 2017-12-11

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Test carried out by: I. Shuryghin

Result: **passed**

Comment: For static thresholds, a variable power supply is used to gradually increase the input voltage while an oscilloscope is used to monitor the status output.
For dynamic on/off timing, a switch is used to turn on/off the input voltages, while an oscilloscope is used to monitor the status output and the input voltage. The results measured meet the requirements.
Please note, 24/48, 96/110, and 48/72 variants are tested and listed in the table below. Variants AAAA, ABAA, and ACAA represent the corresponding variants BAAA, BBAA, and BCAA for this test.

Variant	V_{IH} [V]		V_{IL} [V]		Input voltage [V]	t_{ON} [ms]		t_{OFF} [ms]	
	Required	Measured	Required	Measured		Required	Measured	Required	Measured
7HA 02941 AAAA	≤ 60	47.2	≥ 24	44.8	67.2	1~12	4.4	≤ 0.5	0.38
					77		3.6		0.39
					96		2.64		0.41
					110		2.24		0.42
					120		2.02		0.42
					137.5		1.7		0.44

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7HA 02941 ACAA	≤ 15	12.04	≥ 6	11.35	16.8	1~12	6.3	≤ 0.5	0.30
					24		3.8		0.34
					25.2		3.5		0.35
					30		2.8		0.36
					36		2.2		0.38
					45		1.8		0.39
7HA 02941 ABAA	≤ 30	22.0	≥ 12	21.5	33.6	1~12	6.3	≤ 0.5	0.36
					48		3.7		0.34
					56		3.1		0.32
					72		2.1		0.30
					81		1.9		0.29
					90		1.6		0.26

[HW-IRS_DIM_VI_395][HW-IRS_DIM_VI_396]

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[FVTR_DIM_VI_0027]

Test case: **FVTR_FUNC_14**

Test: Fast Input Circuits

Description: The fast input shall be able to detect input pulses 300 µS or longer. The specifications below shall be met.

Table 12: Fast Input Parameters

V_{IH}	V_{IL}	t_{ON}	t_{OFF}
≤ 15 V	≥ 6 V	≤ 50 µs	≤ 100 µs

Failure criteria: The fast inputs require pulses exceeding 300 µS or the below requirements are not met.

Reference: HW-IRS_DIM_VI_406, HW-IRS_DIM_VI_407, HW-IRS_DIM_VI_408

Date: 2017-08-03

Test carried out by: I. Shuryghin

Result: **passed**

Comment: For static thresholds, a variable power supply is used to gradually increase the input voltage while an oscilloscope is used to monitor the status output.
For dynamic on/off timing, a switch is used to turn on/off the input voltages, while an oscilloscope is used to monitor the status output and the input voltage. The results measured meet the requirements. The test results are included in table below.
Also a function generator is used to supply a pulse train with pulse width 300us +0/-10us to the input; the scope reads the status output with same signal shapes in logic level.

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Variant	V_{IH} [V]		V_{IL} [V]		Input voltage [V]	t_{ON} [us]		t_{OFF} [us]	
	Re-quired	Meas-ured	Re-quired	Meas-ured		Re-quired	Meas-ured	Re-quired	Meas-ured
7HA 02941 BAAA	≤ 15	11.6	≥ 6	11.0	24	≤ 50	12	≤ 100	36
7HA 02941 BCAA		11.85		11.22			3.6		58
7HA 02941 BBAA		11.1		10.6			12.0		32

[HW-IRS_DIM_VI_406][HW-IRS_DIM_VI_407][HW-IRS_DIM_VI_408]

[FVTR_DIM_VI_0028]

Test case: **FVTR_FUNC_15**

Test: Front Panel LEDS

Description: The front panel shall be equipped with 2 dual colour LEDs. Each channel shall have an independently controlled red and green LED controlled by the microcontroller. The LED shall be powered by the microprocessor power supply.

Failure criteria: LEDs not present or not as described.

Reference: HW-IRS_DIM_VI_453, HW-IRS_DIM_VI_454, HW-IRS_DIM_VI_455, HW-IRS_DIM_VI_482

Date: 2017-10-26

Test carried out by: I. Shuryghin

Result: **passed**

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Comment: Front panel LEDs inspected, schematic checked.
There are two dual LEDs on front panel. One red and one green LEDs are controlled by channel A microcontroller and powered by channel A power supply, and another red and another green LEDs are controlled by channel B microcontroller, and powered by channel B power supply.

[HW-IRS_DIM_VI_453][HW-IRS_DIM_VI_454][HW-IRS_DIM_VI_455][HW-IRS_DIM_VI_482]

[FVTR_DIM_VI_0029]

Test case: **FVTR_FUNC_16**

Test: Serial EEPROMs

Description: Each channel shall be provided with an I^2C EEPROM with the size of 1Mbit.

Failure criteria: EEPROMs not present.

Reference: HW-IRS_DIM_VI_477

Date: 2017-11-03

Test carried out by: I. Shuryghin

Result: **passed**

Comment: A 1Mbit I^2C EEPROM with BCN 7FB116690009 is presented for each computer channel (MN2 for channel A, and MN3 for channel B). Memory read and write operations are tested OK with KETEREX USB 910H adapter via the I^2C bus.

[HW-IRS_DIM_VI_477]

[FVTR_DIM_VI_0042]

Test case: **FVTR_FUNC_18**

Test: Hot swap

Description: The card shall operate correctly and not be damaged during hot insertion or removal. Circuitry to limit the inrush current on the 24 V DC power rail should be provided. The microcontroller daughterboard provides inrush limiting on the 5 V and 3.3 V rails.

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Failure criteria:	DIM is damaged or does not function correctly during hotswap. DIM causes other cards in the subrack to malfunction when it is hot inserted or removed.
Reference:	HW-IRS_DIM_VI_526
Date:	2017-10-26
Test carried out by:	I. Shuryghin
Result:	**passed**
Comment:	DIM main board is only using 3.3v supplied by CAN I/O daughter board, the 3.3V is generated from the 5VDC, which has the hot swap function on the CAN I/O daughter board.

[HW-IRS_DIM_VI_526]

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7 Isolation – High Voltage Test

[FVTR_DIM_VI_0031]

Test case: **FVTR_HVT_01**

Test: **High Voltage Test**

Description: The electric strength between the net classes has to be tested with the voltages according to the values declared in [INSUCO_TOP]. The access to the circuit has to be realised via the connectors P1, P2 and P3 on the pins shown in the following table.

Circuit class according to [INSUCO_TOP]	Description	Pin according to [CD_DIM_VI]
C4	Electronics Common	P2 3, 4, 7, 8, 15, 16, 22, 27, 34, 39, 46, 51, 58, 63 P3 3, 4, 7, 8, 15, 16, 22, 27, 34, 39, 46, 51, 58, 63
C4.1	Electronics Replica 1	P2 1, 2, 5, 6, 11, 12, 13, 14, 17, 18, 19, 20, 21, 23, 24, 25, 31, 32, 33, 35, 36, 37, 38, 40, 41, 42, 43, 44, 50, 52, 59, 61, 62, 64
C4.2	Electronics Replica 2	P3 1, 2, 5, 6, 11, 12, 13, 14, 17, 18, 19, 20, 21, 23, 24, 25, 31, 32, 33, 35, 36, 37, 38, 40, 41, 42, 43, 44, 50, 52, 59, 61, 62, 64
C5 (1A)	Input 1 A	P1 d2, z2
C5 (1B)	Input 1 B	P1 d4, z4

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Circuit class according to [INSUCO_TOP]	Description	Pin according to [CD_DIM_VI]
C5 (2A)	Input 2 A	P1 d6, z6
C5 (2B)	Input 2 B	P1 d8, z8
C5 (3A)	Input 3 A	P1 d10, z10
C5 (3B)	Input 3 B	P1 d12, z12
C5 (4A)	Input 4 A	P1 d14, z14
C5 (4B)	Input 4 B	P1 d16, z16
C5 (5A)	Input 5 A	P1 d18, z18
C5 (5B)	Input 5 B	P1 d20, z20
C5 (6A)	Input 6 A	P1 d22, z22
C5 (6B)	Input 6 B	P1 d24, z24
C5 (7A)	Input 7 A	P1 d26, z26
C5 (7B)	Input 7 B	P1 d28, z28
C5 (8A)	Input 8 A	P1 d30, z30
C5 (8B)	Input 8 B	P1 d32, z32
C5	All C5 (*xx*)	P1 all pins in C5 (*xx*)
EGND	Chassis Ground	Faceplate

Table 1 – Circuits for high voltage test

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Circuit 1	Circuit 2	Insulation Type [INSUCO _TOP]	Clearance printed board [mm] internal/external layer	Test voltage [V_{rms}]	AC voltage test OK/ NOK
C4.1	C4.2	SD1	1.8 / 1.2	Not tested (See FVTR_HVT_02)	
C5	C4, C4.1, C4.2, EGND	SD2 (R1)	4.0 / 2.54	2000 V AC, 60 Hz, 1 minute	**OK**
C5 (xA)	C5 (*x*B)	SD 2	3.0 / 2.54	2000 V AC, 50 Hz, 1 minute	**OK**
C5 (1A), (1B)	C5 (2A), (2B),(3A), (3B), (4A), (4B), (5A), (5B), (6A), (6B), (7A), (7B), (8A), (8B)	SD2 (R1)	4.0 / 2.54	2000 V AC, 50 Hz, 1 minute	**OK**
C5 (2A), (2B)	C5 (3A), (3B), (4A), (4B), (5A), (5B), (6A), (6B), (7A), (7B), (8A), (8B)	SD2 (R1)	4.0 / 2.54	2000 V AC, 50 Hz, 1 minute	**OK**
C5 (3A), (3B)	C5 (4A), (4B), (5A), (5B), (6A), (6B), (7A), (7B), (8A), (8B)	SD2 (R1)	4.0 / 2.54	2000 V AC, 60 Hz, 1 minute	**OK**
C5 (4A), (4B)	C5 (5A), (5B), (6A), (6B), (7A), (7B), (8A),	SD2 (R1)	4.0 / 2.54	2000 V AC, 50Hz, 1 minute	**OK**

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Circuit 1	Circuit 2	Insulation Type [INSUCO _TOP]	Clearance printed board [mm] internal/external layer	Test voltage [V_{rms}]	AC voltage test OK/ NOK
	(8B)				
C5 (5A), (5B)	C5 (6A), (6B), (7A), (7B), (8A), (8B)	SD2 (R1)	4.0 / 2.54	2000 V AC, 50Hz, 1 minute	**OK**
C5 (6A), (6B)	C5 (7A), (7B), (8A), (8B)	SD2 (R1)	4.0 / 2.54	2000 V AC, 50Hz, 1 minute	**OK**
C5 (7A), (7B)	C5 (8A), (8B)	SD2 (R1)	4.0 / 2.54	2000 V AC, 50 Hz, 1 minute	**OK**

Table 2 – High voltage test results

Failure criteria: Arcing or current over 1 mA

Reference: HW-IRS_DIM_VI_381, HW-IRS_DIM_VI_459

Date: 2014-08-19

Test carried out by: M. Müller, verified by I. Shuryghin

Result: **passed**

Comment: **Insulation test leakage current is measured less than 0.5mA.**

Test with new boards:
7HA-02919-AAAA Ed.:04 EG 14290 4445 with
7HA-02941-AAAA Ed.:09 EG 14290 4450 and
7HA-02930-AAAA Ed.:07 EG 13411 0781 (Daughterboard)

High Voltage Tester ETL UH36 10167511

NOTE: Since there is no PCB layout change and no key components change on all six board variants, the variant AAAA can be regarded as the test representative for the rest five variants.

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[HW-IRS_DIM_VI_381][HW-IRS_DIM_VI_459]

[FVTR_DIM_VI_0032]

Test case:	**FVTR_HVT_02**
Test:	Layout Constraints
Description:	The constraints from the CAD tool need to be checked. Any violations must be justified to not violate the requirements of [INSUCO_TOP].
Failure criteria:	Constraints not met that violate requirements of [INSUCO_TOP].
Reference:	HW-IRS_DIM_381, HW-IRS_DIM_459
Date:	2014-09-18
Test carried out by:	W. Sheng, verified by I. Shuryghin
Result:	**passed**
Comment:	[ECAD] file REV6 is checked regarding the constraints. The only waived violations related to requirements are the area around the back connectors. When two vital input signals are connected to adjacent pin pairs to the connector, the clearance between the pins is only about 3.4mm, less than the required 4mm. A solution was implemented in the layout to cut a 1mm wide slot along the pin pairs to increase the creepage distance to more than 6.5mm, which meets [INSUCO_TOP]. The actual test results showed that the board meets the insulation voltage requirements as tested in FVTR_HVT_01. Also checked in [ECAD] REV6, the spacing between circuit class electronics (C4) and external inputs (C5) now is 4/2.54mm. Ticket #**TASCM00268496** is resolved.

[HW-IRS_DIM_VI_381][HW-IRS_DIM_VI_459]

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8 Reliability

[FVTR_DIM_VI_0033]

Test case:	**FVTR_REL_01**
Test:	Reliability
Description:	The MTBF, calculated using ALCAREL or MIL-HDBK-217, using the Ground Fixed environment at 45°C shall be greater than 100, 000 hours.
Failure criteria:	MTBF less than 100,000 hours.
Reference:	HW-IRS_DIM_467, HW-IRS_DIM_468
Date:	2013-09-13
Test carried out by:	W. Sheng, verified by I. Shuryghin
Result:	**passed**
Comment:	Calculated using ALCAREL, MTBF = 660, 135 hours for variant AAAA, and 672,241 hours for BAAA variant.

[HW-IRS_DIM_VI_467][HW-IRS_DIM_VI_468]

[FVTR_DIM_VI_0034]

Test case:	**FVTR_REL_02**
Test:	Maintainability
Description:	The MTTR shall be less than 30 minutes.
Failure criteria:	MTTR greater than 30 minutes.
Reference:	HW-IRS_DIM_466
Date:	2012-11-08
Test carried out by:	W. Sheng, verified by I. Shuryghin
Result:	**passed**
Comment:	This board can be easily removed from the subrack for replacement. The MTTR is less than 30 minutes.

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[HW-IRS_DIM_VI_466]

[FVTR_DIM_VI_0035]

Test case:	**FVTR_REL_03**
Test:	Service Life
Description:	The service life of the DIM shall be greater than 20 years.
Failure criteria:	Service life less than 20 years.
Reference:	HW-IRS_DIM_291
Date:	2013-09-13
Test carried out by:	W. Sheng, verified by I. Shuryghin
Result:	**passed**
Comment:	The MTBF is calculated and equivalent to 75 years.

[HW-IRS_DIM_VI_291]

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9 Safety Requirements

[FVTR_DIM_VI_0036]

Test case: **FVTR_SAF_01**

Test: Safety Standards

Description: The design complies with all safety standards defined in [HWADD_TOP]:

- EN 50129 Railway applications - Communication, signalling and processing systems - Safety related electronic systems for signalling [Ed. 2003]
- IEEE 1474.1 Standard for Communications-Based Train Control (CBTC) Performance and Functional Requirements. [Ed. 2004]
- AREMA Communications & Signals Manual of Recommended Practice. [Rev. 2010]

Failure criteria: Requirements of the above standards are not met.

Reference: HW-IRS_DIM_286

Date:

Test carried out by: SAF

Result: **passed**

Comment: Safety was involved in the design reviews and approved the design from safety aspects. Further details should be provided by Safety Experts from Safety department.

[HW-IRS_DIM_VI_286]

[FVTR_DIM_VI_0037]

Test case: **FVTR_SAF_02**

Test: Safety Process

Description: The design process shall comply with a Safety Integrity Level 4.

Failure criteria: Requirements of the above standards are not met.

Reference: HW-IRS_DIM_456

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Date: 2013-09-13

Test carried out by: W. Sheng, verified by I. Shuryghin

Result: **passed**

Comment: The board development followed the process defined in Stages.

[HW-IRS_DIM_VI_456]

[FVTR_DIM_VI_0038]

Test case: **FVTR_SAF_03**

Test: Documentation

Description: All documentation shall be prepared as required by [DOP]

Failure criteria: Documents not available.

Reference: HW-IRS_DIM_475

Date: 2013-09-13

Test carried out by: W. Sheng, verified by I. Shuryghin

Result: **passed**

Comment: All documentation is prepared according to [DOP], reviewed and released in CMCS, archived in PALMA.

[HW-IRS_DIM_VI_475]

[FVTR_DIM_VI_0048]

Test case: ***S* FVTR_SAF_04**

Test: **Inherent fail-safe components SAC usage validation**

Description: Verify all safety components (components with inherent fail-safe properties) for their correct usage according to their safety application conditions.

Failure criteria: Constraints for usage are not considered.

Reference: [SAC_GUIDE], [SAC_MELF0204], [CD_DIM_VI], [EBOM_DIM_VI], [PBATD_DIM_VI]

Date: 2017-08-02

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Test carried out by: I. Shuryghin

Result: **passed**

Comment: According to the [SAC_GUIDE] and [EBOM_DIM_VI] analysis, the following inherent fail-safe components are used on the DIM:

Reference	Type	Value	Purpose
R1_*n*, R2_*n*, R3_*n*, R4_*n*	MELF 0204 1AB 13984 0017 for 7HA02941AAAA;1AB 13984 0004 for 7HA02941BAAA, 7HA02941BBAA, 7HA02941BCAA (FAST) 1AB139840009 for 7HA02941 ACAA 1AB139840014 for 7HA02941ABAA	2.21 kΩ 182 Ω 475 Ω 1.21 kΩ	Trainline protection from short circuit
R5_*n*, R6_*n*, R7_*n*, R8_*n*	MELF 0204 1AB 13984 0018 for 7HA02941AAAA; 1AB 13984 0004 for 7HA02941BAAA, 7HA02941BBAA, 7HA02941BCAA (FAST) 1AB139840010 for 7HA02941ACAA 1AB139840014 for 7HA02941ABAA	2.74 kΩ 182 Ω 562 Ω 1.21 kΩ	Trainline protection from short circuit
R9_*n*, R10_*n,* R11_*n*, R12_*n*	MELF 0204 1AB 13984 0025 for 7HA02941AAAA; 1AB 13984 0007 for 7HA02941BAAA, 7HA02941BBAA,	10 kΩ 332 Ω	Trainline protection from short circuit

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Reference	Type	Value	Purpose
	7HA02941BCAA (FAST) 1AB139840013 for 7HA02941ACAA 1AB139840017 for 7HA02941ABAA	1K Ω 2.21 kΩ	
R13_*n*, R14_*n*	MELF 0204 1AB 13984 0025 for 7HA02941AAAA; 1AB 13984 0014 for 7HA02941BAAA, 7HA02941BBAA, 7HA02941BCAA (FAST) 1AB139840020 for 7HA02941ACAA 1AB139840024 for 7HA02941ABAA	10 kΩ 1.21 kΩ 3.92 kΩ 8.25 kΩ	Device protection (limiting current through electrical safety device)
R15_*n*, R16_*n*	MELF 0204 1AB 13984 0007	332 Ω	Device protection (limiting current through electrical safety device)
R17_*n*, R18_*n*, R19_*n*, R20_*n*	MELF 0204 1AB 13984 0013	1 kΩ	Internal electronics channel separation

The analysis against the conditions [SAC_MELF0204] is shown below:

Requirement	Value	Required Evidence	Evidence	Compliance
5.1.1.a) Maximum operating temperature / Permissible film temperature	+ 125 °C	*1.Check IRS, if the maximum operating temperature fit to the required temperature range of the application (board / device)* *2. Show with temperature*	1. [HW-IRS_DIM] states the maximum board level operating temperature is 85°C. 2. See [QTR] Due to derating the power limit	Yes

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Requirement	Value	Required Evidence	Evidence	Compliance
		test of the board / device, that this condition will be fulfilled (e. g. by measurement of environmental temperature in the surrounding of the component). Consider the derating of the resistor above 70°C.	at 85°C is 175 mW.	
5.1.1.b) Minimum operating temperature	- 55 °C	*1.Check IRS, if the minimum operating temperature fit to the required temperature range of the application (board / device)* *2. Show with temperature test of the board / device, that this condition will be fulfilled. Because of heating in operation (increase of component temperature), a measurement of environmental temperature in the surrounding of the component is not needed.*	1. [HW-IRS_DIM] states the maximum board level operating temperature is -40°C. 2. See [QTR]	Yes
5.1.1.c) Pollution degree	≤ PD2	*Check the clearance and creepage distances in the finalized layout and give an according statement.*	Creepage and clearance distances are taken from [INSUCO_TOP] which uses PD2 for external layers and PD1 for internal layers.	Yes
5.1.2.a) Maximum operating voltage	200 V_{eff}	*Check schematic for possible maximum voltage, also in case of error.*	R1_n, R2_n, R3_n, R4_n: 137.5V for 7HA02941AAAA; 24V for 7HA02941BAAA, 7HA02941BBAA, 7HA02941BCAA (FAST) 45V for 7HA02941ACAA 90V for 7HA02941ABAA R5_n, R6_n, R7_n, R8_n: 137.5V for 7HA02941AAAA; 24V for 7HA02941BAAA, 7HA02941BBAA, 7HA02941BCAA (FAST)	Yes

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Requirement	Value	Required Evidence	Evidence	Compliance
			45V for 7HA02941ACAA 90V for 7HA02941ABAAR9_n, R10_n, R11_n, R12_n: 137.5V for 7HA02941AAAA; 24V for 7HA02941BAAA, 7HA02941BBAA, 7HA02941BCAA (FAST); 45V for 7HA02941ACAA; 90V for 7HA02941ABAA; R13_n, R14_n: 3.3V R15_n, R16_n: 3.3V R17_n, R18_n, R19_n, R20_n: 3.3V	
5.1.2.b) Limiting rated power P_{70}	0.25 W	*Check schematic for possible maximum power rating: consider the derating above 70°C.* *Calculate power rating with possible maximum voltage also in case of error.* *The limited power rating must never be exceeded.*	Due to derating the power limit at 85°C is 175 mW R1_n, R2_n, R3_n, R4_n: 47mW for 7HA02941AAAA; 50mW for 7HA02941BAAA, 7HA02941BBAA, 7HA02941BCAA (FAST); 58mW for 7HA02941ACAA; 42mW for 7HA02941ABAA;R5_n, R6_n, R7_n, R8_n: 58mW for 7HA02941AAAA; 50mW for 7HA02941BAAA, 7HA02941BBAA, 7HA02941BCAA (FAST); 69mW for 7HA02941ACAA; 42mW for 7HA02941ABAA; R9_n, R10_n, R11_n, R12_n: 106mW for 7HA02941AAAA; 91mW for 7HA02941BAAA, 7HA02941BBAA, 7HA02941BCAA (FAST); 122mW for 7HA02941ACAA 166mW for 7HA02941ABAA; R13_n, R14_n: 1mW for 7HA02941AAAA; 9mW for 7HA02941AAAA, 7HA02941BBAA, 7HA02941BCAA (FAST); 3mW for 7HA02941ACAA; 1mW for 7HA02941ABAA;	Yes

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Requirement	Value	Required Evidence	Evidence	Compliance
			R15_n, R16_n: 33mW R17_n, R18_n, R19_n, R20_n: 11mW	
5.1.2.c) Isolation voltage	300 V (DC or AC peak 1 min); 75 V (DC or AC continuous)	*Check of isolation voltage not possible after board/device production, therefore:* *Perform optical check of final layout, that there are no copper tracks below resistor body (between footprints).*	Checked layout. No metal under resistors (between pads): R1_n, R2_n, R3_n, R4_n, R5_n, R6_n, R7_n, R8_n, R9_n, R10_n, R11_n, R12_n, R13_n, R14_n, R15_n, R16_n, R17_n, R18_n, R19_n, R20_n	Yes
5.1.2.d) Clearance and creepage between caps	On the assembled board according to Mü 8004, Teil 37200	*For the maximum operating voltage of 200 V_{eff} (according to 5.1.2.a)) and <= PD2 the clearance and creepage distance is according to Mü 8004, Teil 37200 always fulfilled, if the over voltage in the circuit is lower than 1,5 kV impulse voltage (impulse voltage type 1,2/50 µs).* *CID has to check, that in the circuit the maximum possible impulse voltage is below the allowed value.*	The impulse over voltage UNI on DIM board for each resistor is below 1,5 kV.	Yes

All the SACs for the safety components are fulfilled.

[HW-IRS_DIM_VI_458]

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10 Environmental and EMC Tests

Environmental and EMC requirements are tested at the subrack or system level and are covered in [QTS-EI_COMM-IO_SR] and [QTR-PRD-HW_IO-EXP_SR].

Reference: HW-IRS_DIM_415, HW-IRS_DIM_430, HW-IRS_DIM_433, HW-IRS_DIM_435, HW-IRS_DIM_437, HW-IRS_DIM_439, HW-IRS_DIM_441, HW-IRS_DIM_443, HW-IRS_DIM_445

[FVTR_DIM_VI_0049]

Test case:	**FVTR_ENV_01**
Test:	Module temperature measurement
Description:	DIM VI board does not have sensors but the temperature measurements in the range -45C to +100C with an accuracy of 2% FSR – or better are done by the daughter card.
Failure criteria:	The module has no temperature sensors.
Reference:	HW-IRS_DIM_525
Date:	2013-07-18
Test carried out by:	D. Praisler
Result:	**passed**
Comment:	Tested with the module in the rack.

[HW-IRS_DIM_VI_525]

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11 Open defects

None.

END OF DOCUMENT

THALES GROUP INTERNAL

Entity Identi-fier	Document Identifier	DTC	Revision	Customer ID with Revision
15900~9950	7HA-02941-AAAA-980EN	980	009	N/A
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